

AFREEN KHAN

Assistant Professor, Department of Computer Application
Faculty of Engineering & IT

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[Google Scholar](#) | [ORCID](#) | [ResearchGate](#) | [LinkedIn](#)

Education

- **Doctor of Philosophy**, Computer Science
Aligarh Muslim University, Aligarh, Uttar Pradesh, India, August 2022
- **Masters of Computer Science and Application**
Aligarh Muslim University, Aligarh, Uttar Pradesh, India, November 2016
- **Bachelor of Science**, Computer Application Aligarh Muslim
University, Aligarh, Uttar Pradesh, India, June 2013

Work Experience

- **Assistant Professor**, Department of Computer Application, Faculty of Engineering & IT, Integral University, Lucknow, India, since September 2022
- **Visiting Lecturer**, Yesbud University (<https://yesbud.online/>), Jan-Mar 2023
- **Guest Instructor**, CSDCP, Women's College, Aligarh Muslim University, Aligarh, India, May-Sep 2022
- **Senior Research Associate**, Company: On My Own Technology Pvt. Ltd. (OMOTEC), Mumbai (WFH), Dec 2021 - Mar 2022
- **Research Associate**, Company: SoftNice India Pvt. Ltd. [Department: Microsoft], Aligarh and Noida, Mar-Sep 2021

Research Interests

- Machine Learning and Deep Learning for Clinical Decision Support and Disease Prediction
- Electronic Health Record (EHR) Mining, Time-Series Analysis, and Statistical Modeling
- Natural Language Processing for Clinical Narratives, Social Determinants of Health, and LLM Applications
- Big Data Analytics and AI Applications in Public Health, Epidemiology, and Health Equity
- Genomic and Multi-Omics Data Science for Early Detection of Neurodegenerative Diseases
- Climate-Health Informatics and Environmental Data Science for Population Health Monitoring
- Analytics-Enhanced e-Learning, Blended Learning, and Education Informatics

Honours and Awards

- Maulana Azad National Fellowship for M.Phil/Ph.D. student, University Grants Commission, Govt. of India, Apr 2019 – Aug 2022
- PhD Scholarship, Aligarh Muslim University, Aligarh, India, Dec 2017 – Mar 2019
- Best Poster Presentation Award, National Seminar on Cashless Economy, Aligarh Muslim University, Aligarh, India, Mar 2017
- Post Graduate Merit Scholarship, Aligarh Muslim University, Aligarh, India, Jul 2015 – Jun 2016
- Merit-cum-Means Scholarship, Govt. of Uttar Pradesh, India, Mar 2014 – Mar 2015
- Top Dissertation Holder (MCA), Aligarh Muslim University. November 2016

Research Projects and Funding

1. **AI-Driven framework for building recovery resilience in severe mental illness through social determinants across urban & rural**
Principal Investigator, Funded by Integral Centre of Excellence for Interdisciplinary Research, Integral University, Lucknow, India. Total grant amount: Rs. 2 Lakhs, Apr 2025 – Mar 2027
2. **Mental Health Impact of the COVID-19 Pandemic on the Rural Population: A Community Based Cross-Sectional Study from North India**
Research Contributor, Funded by Integral Institute of Medical Sciences and Research (IIMSR), Integral University, Lucknow, India. Total grant amount: Rs. 1.50 Lakhs (Supervisors: Dr Deepak Chopra, Dr Samreen Khan, Dr Siraj Ahmed, Mr. Rafi and Dr Afreen Khan), May 2021 – Apr 2023

Peer-Reviewed Journal Articles

Google scholar profile (h-index: 9, i10-index: 9; total citations: 291):

<https://scholar.google.com/citations?user=1nYfL-MAAAAJ&hl=en>

1. **Khan A**, Zubair S, Ali I. ADNeuroNet: A Neuroevolution-based Neural Network Algorithm for the Diagnosis of Neurodegenerative diseases. *Neural Computing and Applications*, 37, 8593–8627, 2025.
2. **Khan A.**, Zubair S, Shuaib M, Sheneamer A, Alam S, Assiri B. Development of a Robust Parallel and Multi-Composite Machine Learning Model for Improved Diagnosis of Alzheimer's Disease: Correlation with Dementia-Associated Drug Usage and AT(N) Protein Biomarker. *Frontiers in Neuroscience*, 18, 2024.
3. Parvez S, Zubair S, **Khan A**. A Hybrid Approach for Weak Learners Utilizing Ensemble Technique for Alzheimer's Disease Prognosis. *Indian Journal of Science and Technology*, 16(32), 2518-2533, 2023.
4. **Khan A**, Zubair S. Development of a Three Tiered Cognitive Hybrid Machine Learning Algorithm for Effective Diagnosis of Alzheimer's Disease. *Journal of King Saud University - Computer and Information Sciences*, 34(18A), 8000-8018, 2022.
5. **Khan A**, Zubair S, Khan S. A systematic analysis of assorted machine learning classifiers to assess their potential in accurate prediction of dementia. *Arab Gulf Journal of Scientific Research*, 40(1), 2022.

6. **Khan A**, Zubair S, Khalique N, Khan S. An exploratory study to find the early trend and pattern recognition of Covid-19 infection in India: A Severity model based prediction. *Indian Journal of Community Health*, 33(20< 299-303, 2021.
7. **Khan A**, Zubair S, Khan S. Comprehensive Performance Analysis of Neurodegenerative disease Incidence in the Females of 60-96 year Age Group. *Advances in Distributed Computing and Artificial Intelligence Journal*, 10(2), 183-196, 2021.
8. **Khan A**, Zubair S. An Improved Multi-Modal based Machine Learning Approach for the Prognosis of Alzheimer's Disease. *Journal of King Saud University - Computer and Information Sciences*, 34(6A), 2688-2706, 2020.
9. **Khan A**, Zubair S. Longitudinal Magnetic Resonance Imaging as a Potential Correlate in the Diagnosis of Alzheimer Disease: Exploratory Data Analysis. *JMIR Biomedical Engineering*, 5(1), 1-13, 2020.
10. **Khan A**, Zubair S. Usage of Random Forest Ensemble Classifier based Imputation and its potential in the diagnosis of Alzheimer's Disease. *International Journal of Scientific and Technology Research*, 8(12), 2019.
11. Zubair S, Sabri MAI, **Khan A**. Correlation among Similarity Measurements for Collaborative Filtering Techniques: An Improved Similarity Metric. *Journal of Analysis and Computation*, 2019.
12. **Khan A**, Zubair S. Machine Learning Tools and Toolkits in the Exploration of Big Data. *International Journal of Computer Sciences and Engineering*, 6(12), 570-575, 2018.
13. Bokhari MU, **Khan A**. The NoSQL Movement. *IOSR Journal of Computer Engineering*, 18(6), 06-12, 2016.

Refereed Book Chapters

1. Vakil MK, **Khan A**, Rahman A. Analysing the Academic Performance of Students Under Varying Climatic Conditions, in Muzammil M, Khan AA, Hasan F, (eds) *Ergonomics for Improved Productivity*, Springer, Singapore, 2022.

Conference Proceedings

1. Camran A, **Khan A**. An Automated Framework for Extracting Medical Entities from Unstructured EHRs Using Biomedical NER Models. *16th International IEEE Conference on Computing, Communication and Networking Technologies*, IIT Indore, Madhya Pradesh, 06-11 Jul, 2025.
2. Camran A, **Khan A**. The Role of LLMs for Optimizing Clinical Insights in Text Summarization. *5th International Conference on Intelligent Vision and Computing*, ICFAI University, Dehradun, 13-14 Jun, 2025. Accepted for publication in: Springer Lecture Notes in Networks and Systems (LNNS).
3. **Khan A**. Nilearn-based Multimodal Fusion Analysis for Integrating Brain Image Analysis. *14th International Conference on Computing Communication and Networking Technologies*, IEEE Conference Proceedings, IIT Delhi, India, Jul 2023.
4. **Khan A**, Zubair S, Khan S. An Epidemiological-based Regression Analysis of Alzheimer's disease and Mild Cognitive Impairment Converts in the Female Population. *IEEE EUROCON-2021 - 19th International Conference on Smart Technologies*, IEEE Conference Proceedings, Lviv, Ukraine, Jul 2021.

5. **Khan A**, Zubair S, Khan S. Prospectives of Big Data Analytics and Explainable Machine Learning in Identification of Probable Biomarkers of Alzheimer's disease. *The 1st International Conference on Computing and Machine Intelligence*, Istanbul, Turkey, 19-20 Feb, 2021. Published in: Manchester Journal of Artificial Intelligence & Applied Sciences.
6. **Khan A**, Zubair S. A Machine Learning Based Robust Approach to Identify Dementia Progression Employing Dimensionality Reduction in Cross Sectional MRI Data. *The First International Conference of Smart Systems and Emerging Technologies*, IEEE Conference Proceedings, Prince Sultan University, Riyadh, Saudi Arabia, Nov 2020.
7. **Khan A**, Zubair S. Expansion of Regularized Kmeans Discretization Machine Learning Approach in Prognosis of Dementia Progression. *11th International Conference on Computing, Communication and Networking Technologies*, IEEE Conference Proceedings, IIT, Kharagpur, West Bengal, India, Jul 2020.
8. **Khan A**, Zubair S, Sabri MAI. An Improved Pre-processing Machine Learning Approach for Cross-Sectional MR Imaging of Demented Older Adults. *The First International Conference of Intelligent Computing and Engineering (ICOICE-2019)*, IEEE Conference Proceedings, Hadhramout University, Yemen, Dec 2019.
9. **Khan A**, Vakil MK, Bin MSA. Proposed 4SM Model for Indian Education System Implementing Big Data Analytics and Blended Learning Methods. *3rd International Conference for Convergence in Technology (I2CT)*, IEEE Conference Proceedings, Pune, India, Dec 2018.
10. **Khan A**, Bokhari MU. Data Science in Action: Key to Cybersecurity. *5th International Conference on Computing for Sustainable Global Development*, IEEE Conference Proceedings, BVICAM, Delhi, Mar 2018.
11. Bokhari MU, **Khan A**, Critical Review on Threat Model of Various NoSQL Databases. *4th International Conference on Computing for Sustainable Global Development*, IEEE Conference Proceedings, BVICAM, Delhi, 01-03 Mar, 2017.

International Conferences: Oral Presentations

1. **Khan A**, Bokhari MU. Big Data Analytics as a Tool for Reducing Ergonomics Risk. *15th International Conference on Humanizing Work and Work Environment*, Zakir Hussain College of Engineering and Technology, Aligarh Muslim University, Aligarh, India, Dec 2017.
2. **Khan A**, Bokhari MU. Ergonomic Survey of Computer Science Students: Challenges at Workplace. *15th International Conference on Humanizing Work and Work Environment*, Zakir Hussain College of Engineering and Technology, Aligarh Muslim University, Aligarh, India, Dec 2017.

National Conference: Oral Presentation

1. **Khan A**. Cashless Economy in the Age of Big Data. National Conference on Cashless Economy: Opportunities and Challenges, Aligarh Muslim University, Aligarh, India, Apr 2017.

International Symposium: Oral/Poster Presentation

1. **Khan A**, Khan S. Alzheimer's disease and mild cognitive impairment converts in the female population: An epidemiological analysis. *18th Singapore Public Health & Occupational Medicine Conference*, SingHealth, Singapore, 23-24 Sep 2024.

National Symposium: Oral/Poster Presentation

1. **Khan A**, Zubair S. An Improved Model on Climate Change and its Impact on Health employing Machine Learning Methods. *National Symposium on Biodiversity and Sustainable Development*, Department of Zoology, Aligarh Muslim University, Aligarh, India, Nov 2019.

Participation in International/ National Conferences/ Symposia/ Seminars

1. **Indo-German Symposium**, *Recent Advances in Machine Translation*, NIT Assam, India, Mar 2021.
2. *13th International Symposium on Health Informatics and Bioinformatics*, Turkey, Oct 2020.
3. *International Conference on Emerging Trends in Healthcare Technology*, IIT Kharagpur, India, Aug 2020.
4. **Symposium**, *Quantum Information & Computation*, IIIT Hyderabad, India, Jul 2020.
5. **National Seminar**, *Data Science and Its Applications*, TCC, Maharashtra, India, Jul 2020.
6. **National Conference**, *Biomonitoring*, AISECT College, India, Jun 2020.
7. **International Seminar**, *Future Perspectives for COVID-19*, IEEE, GIET University, India, May 2020.

Advising and Mentoring

- Currently supervising 2 PhD students, 4 MCA students and 8 BCA students
- 9 MCA students and 10 BCA students graduated so far; 2 of these students were external students, i.e. registered in other institutions/universities

Teaching and Training

1. **Lecturer, CA552: Machine Learning with Python Programming** (4 credits, 78 graduate students), Integral University, Lucknow, Autumn 2024 – 2025 (2-year average rating 94/100)
2. **Mentor & Lecturer, CA453: Fundamentals of Computer and C Programming** (4 credits, 145 graduate students), Integral University, Lucknow, Autumn 2023 – 2025 (3-year average rating 93/100)
3. **Lecturer, CA701: Software Engineering** (4 credits, 10-15 PhD students), Integral University, Lucknow, Spring 2024 – 2025 (2-year average rating 90/100)
4. **Mentor & Lecturer, CA466: Data Science** (4 credits, 145-150 graduate students), Integral University, Lucknow, Spring 2023 – 2025 (3-year average rating 94/100)
5. **Lecturer, CA335: Fundamentals of Machine Learning with Python** (4 credits, 75 undergraduate students), Integral University, Lucknow, Spring 2025
6. **Lecturer, CA561: Internet of Things** (4 credits, 74 graduate students), Integral University, Lucknow, Autumn 2024
7. **Mentor & Lecturer, CA328: Data Analytics and Visualization** (4 credits, 89 undergraduate students), Integral University, Lucknow, Autumn 2024
8. **Lecturer, CA576: Blockchain Computing** (4 credits, 150 graduate students), Integral University, Lucknow, Spring 2024
9. **Lecturer, CA307: Image Processing** (4 credits, 74 undergraduate students), Integral University, Lucknow, Autumn 2023
10. **Mentor & Lecturer, CA580: Big Data Visualization** (4 credits, 76 graduate students), Integral University, Lucknow, Spring 2023

11. **Lecturer, CA315: Cyber Law and Internet Security** (4 credits, 70 undergraduate students), Integral University, Lucknow, Spring 2023
12. **Lecturer, CA113: Cyber Crime and Cyber Law** (4 credits, 78 undergraduate students), Integral University, Lucknow, Spring 2023
13. **Lecturer, CA110: Computer Fundamentals and C Programming** (4 credits, 72 undergraduate students), Integral University, Lucknow, Autumn 2022
14. **Lecturer, F010103T: Computer Applications** (4 credits, 68 BBA students), Integral University, Lucknow, Autumn 2022

Administrative Services

- **Convener, Women Grievance Redressal Cell**, Integral University, Lucknow, since 2023
- **Member, Departmental Research and Publication Cell (DRPC)**, Department of Computer Application, Integral University, Lucknow, since 2025
- **Member, Organizing Committee, International Conference on Advances in Science, Engineering, and Technology (ICASET - 2024)**, Integral University, Lucknow, 30–31 Jul 2024
- **Convener, Departmental Placement Committee**, Department of Computer Application, Integral University, Lucknow, since 2023
- **Member, Surprise Proctorial Duty**, Integral University, Lucknow, since 2023
- **Member Secretary, Innovative Teaching, Learning & Research Committee**, Department of Computer Application, Integral University, Lucknow, since 2023
- **Criteria Member, Departmental Quality Assurance Cell (DQAC) - Criteria-V, Student Support & Progression**, Department of Computer Application, Integral University, Lucknow, since 2023
- **Member, Accreditation & Ranking Committee (ARC)**, Department of Computer Application, Integral University, Lucknow, since 2024
- **Member, Curriculum Review and Development Committee (CRDC)**, Department of Computer Application, Integral University, Lucknow, since 2023
- **Member, Departmental Examination Committee (Theory)**, Department of Computer Application, Integral University, Lucknow, since 2023
- **Member, Student Project Monitoring & Evaluation Committee**, Department of Computer Application, Integral University, Lucknow, since 2023
- **Member, Teaching, Learning, and Evaluation National Assessment and Accreditation Council (NAAC)**, Department of Computer Application, Integral University, Lucknow, 2022-2023

Trainings, Workshops and Certifications

1. **Summer School, Evolutionary Algorithms and Machine Learning**, Department of Mathematics & Computing, NIT Jalandhar, India, 16-20 Jun 2025.
2. **Faculty Development Program, AI in Medical Imaging and Diagnostics: Current Trends and Challenges**, Malaviya National Institute of Technology, Jaipur, India, 01–12 May 2025.
3. **Winter School, Computational Intelligence and Generative AI**, IEEE CIS Chapter, IIT Indore, India, 08-09 Jan 2025.
4. **Online Course Instructor Training Program**, UTEL University, Online University of Mexico, issued 2023.
5. **Training, Geospatial Data Science for Cancer Control and Prevention**, Zhu Family Center for Global Cancer Prevention, Harvard T.H. Chan School of Public Health, Virtual, Jun – Oct 2022.

6. **Faculty Development Program**, *Optimization Techniques for Deep Learning*, Malaviya National Institute of Technology, Bhopal, India, Jan 2022.
7. **Faculty Development Program**, *Biomedical Devices for Precision Health*, IIT Tirupati, India, Jan 2022.
8. **Short Term Course**, *Machine Learning in Healthcare*, National Institute of Technology, Raipur, India, Oct 2020.
9. **Short Term Course**, *Modern Optimization Techniques*, National Institute of Technology, Arunachal Pradesh, India, Aug 2020.
10. **Summer School**, *IEEE Artificial Intelligence Summer School*, Canberra, Virtual, Dec 2020.
11. **Winter School**, *Fog/Edge Computing*, IIT Kanpur, India, Dec 2020.
12. **Training**, *Machine Learning/ Big Data and Hadoop/ Data Science*, National Institute of Electronics & Information Technology (NIELIT), New Delhi, India, issued Jun 2020.

Session Chair/ Invited Lectures

- **Session Chair**, *5th International Conference on ICT for Digital, Smart and Sustainable Development*, Department of CSE, SEST, Jamia Hamdard, New Delhi, India, 22-23 Apr 2025.
- **Invited Speaker and Workshop Facilitator**, *3-day Workshop on Data Science: How To Get Started*, Integral University, Lucknow, India, 30 May - 1 Jun, 2023.
- **Invited Speaker and Workshop Facilitator**, *5-days Skill-based Training programme on Data Analysis and Visualization with Python*, CSDCP, Women's College, Aligarh Muslim University, Aligarh, India, 30 Jan - 3 Feb, 2023.
- **Invited Speaker**, *Artificial Intelligence in Medical Science*, 1-day Continuing Medical Education (CME) Program, Integral Institute of Medical Sciences & Research, Integral University, Lucknow, India, 21 Dec 2022.
- **Invited Speaker**, *Cyber Crimes and Safety*, in observance of Cyber Jaagrookta Divas, 1-day Awareness Program, CSDCP, Women's College, Aligarh Muslim University, Aligarh, India, 1 Jun 2022.

Professional Membership

- Member, *IEEE Computer Society*, since 2013
- Member, *IEEE Women in Engineering*, since 2013
- Life member, *International Association of Engineers*, since 2019